

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

Answer

- A. $p + r + s$
- B. $20(p + r + s)$
- C. $\frac{prs}{pr+ps+rs}$
- D. $\frac{prs}{20p+20r+20s}$

Correct Answer: C

Rationale

Choice C is correct. Multiplying each side of the given equation by $\frac{1}{20}$ yields $\frac{1}{20}\left(\frac{20}{p}\right) = \frac{1}{20}\left(\frac{20}{q} - \frac{20}{r} - \frac{20}{s}\right)$. Distributing $\frac{1}{20}$ on each side of this equation yields $\frac{20}{20p} = \frac{20}{20q} - \frac{20}{20r} - \frac{20}{20s}$, or $\frac{1}{p} = \frac{1}{q} - \frac{1}{r} - \frac{1}{s}$. Adding $\frac{1}{r} + \frac{1}{s}$ to each side of this equation yields $\frac{1}{s} + \frac{1}{r} + \frac{1}{p} = \frac{1}{q}$. Multiplying $\frac{1}{s}$ by $\frac{pr}{pr}$, $\frac{1}{r}$ by $\frac{ps}{ps}$, and $\frac{1}{p}$ by $\frac{rs}{rs}$ yields $\frac{pr}{prs} + \frac{ps}{prs} + \frac{rs}{prs} = \frac{1}{q}$, which is equivalent to $\frac{pr+ps+rs}{prs} = \frac{1}{q}$. Since $\frac{pr+ps+rs}{prs} = \frac{1}{q}$, and it's given that p , q , r , and s are positive, it follows that the reciprocals of each side of this equation are also equal. Thus, $\frac{prs}{pr+ps+rs} = \frac{q}{1}$, or $\frac{prs}{pr+ps+rs} = q$. Therefore, $\frac{prs}{pr+ps+rs}$ is equivalent to q .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.